

USER-FRIENDLY AUTONOMOUS PARCEL DELIVERY SYSTEM WITH ADAPTIVE NAVIGATION



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CONFIDENTIALITY STATEMENT

This report is submitted solely for the purpose of Final Year Project academic evaluation. The project has been developed under the supervision of EIOTAI and is subject to a Non-Disclosure Agreement (NDA).

DECLARATION

It is declared that this is an original piece of our work, except where otherwise acknowledged in text and references. This work has not been submitted in any form for another degree or diploma at any university or other institution for tertiary education and shall not be submitted by us in future for obtaining any degree from this or any other University or Institution.

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CERTIFICATE OF APPROVAL

It is certified that the project titled “**USER-FRIENDLY AUTONOMOUS PARCEL DELIVERY SYSTEM WITH ADAPTIVE NAVIGATION**” carried out by Muhammad Ali Hassan, Reg. No. BEE223007, Aitzaz Arshad, Reg. No. BEE223010, and Gulfam Haider, Reg. No. BEE223020, under the supervision of Engr. Suleman Khan and co-supervision of Dr. Muhammad Atif, Capital University of Science & Technology, Islamabad, is fully adequate, in scope and in quality, as a final year project for the degree of BS Electrical Engineering.

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ABSTRACT

The increasing demand for automation in logistics and indoor service environments has accelerated research in autonomous mobile robot navigation systems. This project presents the design and implementation of a simulation-based autonomous indoor parcel delivery system developed using Robot Operating System 2 (ROS 2) and the Gazebo simulation platform. The entire system is modeled, implemented, and evaluated in a virtual environment without the use of physical hardware. The primary objective of the project is to develop a reliable navigation framework that enables a differential-drive mobile robot to autonomously plan and execute parcel delivery tasks within structured indoor environments such as university buildings, offices, and hospitals. An Improved A* (IA*) algorithm is employed for global path planning to generate collision-free routes on occupancy grid maps. To address the discontinuities and sharp turns produced by grid-based planners, spline-based path smoothing techniques are implemented to generate continuous and kinematically feasible trajectories. A Model Predictive Control (MPC) strategy is designed to ensure accurate trajectory tracking while respecting motion constraints and maintaining system stability. The system is fully integrated within the ROS 2 architecture, utilizing standard communication mechanisms for localization, mapping, navigation, and control. A web-based interface is developed to allow users to assign delivery goals and monitor robot behavior in real time during simulation. Performance evaluation in multiple simulated indoor scenarios demonstrates reliable navigation, effective obstacle avoidance, and smooth trajectory execution. This work provides a scalable software foundation for autonomous indoor delivery systems and establishes a framework for future hardware deployment and real-world implementation.

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LIST OF ACRONYMS/ABBREVIATIONS

AI	Artificial Intelligence
A*	A-Star Path Planning Algorithm
AMCL	Adaptive Monte Carlo Localization
API	Application Programming Interface
CPU	Central Processing Unit
DC	Direct Current
GPS	Global Positioning System
GUI	Graphical User Interface
IA*	Improved A-Star Algorithm
IMU	Inertial Measurement Unit
IoT	Internet of Things
LiDAR	Light Detection and Ranging
MPC	Model Predictive Control
Nav2	Navigation 2 Framework
PID	Proportional–Integral–Derivative
PSO	Particle Swarm Optimization
ROS	Robot Operating System
RRT	Rapidly-Exploring Random Tree
SLAM	Simultaneous Localization and Mapping
UART	Universal Asynchronous Receiver Transmitter
Wi-Fi	Wireless Fidelity

Chapter 1

INTRODUCTION

1.1 Overview

This project presents a simulation-based autonomous indoor parcel delivery system developed using the Robot Operating System 2 (ROS 2) and the Gazebo simulation environment. The aim of the system is to model an intelligent mobile robot that can autonomously navigate indoor spaces such as university buildings, office corridors, and institutional facilities. All development, testing, and evaluation are carried out entirely in simulation, without the use of physical hardware.

The system focuses on core navigation tasks including mapping, localization, path planning, trajectory smoothing, and obstacle avoidance. An Improved A* (IA*) algorithm is used for global path planning to generate collision-free routes on occupancy grid maps. However, since grid-based planners typically produce paths with sharp turns that are difficult for differential-drive robots to follow, a spline-based path smoothing technique is applied to generate smooth and kinematically feasible trajectories. To accurately follow these trajectories, a Model Predictive Control (MPC) approach is implemented, allowing the robot to track planned paths while respecting motion and stability constraints.

The entire system is integrated within the ROS 2 framework, using standard communication mechanisms such as topics, services, and action servers to coordinate navigation and control. In addition, a web-based control interface is developed to allow users to assign delivery tasks, set goal locations, and monitor the robot's behavior in real time during simulation. By adopting a simulation-only approach, the project enables safe and efficient testing of autonomous navigation algorithms while providing a strong foundation for future hardware implementation and real-world deployment.

1.2 Project idea

This project focuses on the development of a simulation-based autonomous indoor parcel delivery system using ROS 2. A virtual mobile robot is designed to navigate indoor environments such as university buildings and office corridors, delivering parcels without human intervention. The system emphasizes core navigation tasks including path

planning, obstacle avoidance, and smooth motion control, all implemented and evaluated within a simulated environment. A simple web-based interface is provided for assigning delivery goals and monitoring system operation.

1.3 Purpose of the project

The purpose of this project is to design and evaluate a simulation-based autonomous indoor parcel delivery system using ROS 2. The project aims to develop a reliable navigation framework that allows a mobile robot to autonomously move within indoor environments while safely avoiding obstacles and following smooth, feasible paths.

By implementing and testing the system entirely in simulation, the project focuses on improving key aspects of autonomous navigation such as path planning, trajectory smoothing, and control performance. The work provides a practical software-based foundation that can support future hardware implementation and further research in autonomous indoor delivery systems.

1.4 Motivation and Need of the Project

The motivation for this project arises from the increasing demand for automation and contactless services, particularly in the post-pandemic era. Many organizations aim to minimize the physical effort involved in repetitive and labor-intensive tasks while maintaining high service quality. Shortages of labor, increasing operational costs, and the need for 24/7 service further highlight the importance of autonomous delivery systems. Moreover, in environments such as hospitals, industrial facilities, and emergency zones, human-based delivery can be risky and unsafe. Autonomous robots can operate in such conditions efficiently while supporting safer workplaces. These needs are addressed in this project by offering an effective, dependable, and user-friendly autonomous indoor parcel delivery solution designed for structured environments.

1.4.1 Project Specifications

This section presents the technical and functional specifications of the proposed User-Friendly Autonomous Parcel Delivery System with Adaptive Navigation. The specifications define the system environment, navigation framework, control strategy, software tools, and safety mechanisms used in the design and implementation.

Table 1.1: System Specifications

Category	Specification
System Type	Software-based autonomous indoor parcel delivery system
Operating Environment	Indoor structured environments (hospitals, offices, universities)
Robot Platform (Simulated)	Differential-drive mobile robot (TurtleBot3 Waffle model in ROS 2/Gazebo)
Navigation Framework	Custom-developed navigation stack (in-house path planning, smoothing, and control algorithms) integrated within the ROS 2 architecture
Path Planning Algorithm	Improved A* (IA*)
Path Smoothing Method	Particle Swarm Optimization (PSO)
Localization Method	SLAM-based localization (LiDAR + odometry)
Obstacle Avoidance	Real-time dynamic obstacle avoidance
Motion Control	Model Predictive Control (MPC)
Map Representation	2D Occupancy Grid Map
Simulation Tools	Gazebo + RViz
Programming Languages	Python (ROS 2 & Flask), Dart (Flutter UI), SQL
Development Platform	Ubuntu Linux with ROS 2

The system is designed as a fully software-based autonomous delivery platform, allowing development and testing in a simulated indoor environment before any hardware deployment. Navigation, mapping, and task execution are handled by a custom-developed navigation stack built entirely in-house and integrated within the ROS 2 architecture, rather than relying on the standard Nav2 stack. An Improved A* algorithm is used for global path planning, while Bezier curve interpolation ensures smooth and kinematically feasible trajectories for a differential-drive robot. SLAM-based localization with LiDAR and odometry provides accurate pose estimation, and real-time obstacle avoidance allows safe operation in dynamic environments. Motion control is achieved using a self-designed Model Predictive Controller (MPC), which improves tracking accuracy and stability. A 2D occupancy grid map is used for environment representation. The system is implemented using Python for ROS 2 and Flask backend services, and Dart with Flutter for the user interface. Safety features such as obstacle inflation, collision avoidance, and emergency stop mechanisms ensure reliable and secure operation in shared indoor spaces.

1.5 Project Objectives

1.5.1 General Objectives

The primary objective of this project is to design and implement an autonomous parcel delivery system capable of navigating indoor environments safely and efficiently. The system aims to minimize human involvement, improve delivery accuracy, and enhance operational efficiency through intelligent navigation and control.

1.5.2 Specific Objectives

The specific objectives of this project are:

- To design an autonomous mobile robot capable of safe indoor navigation.
- To develop adaptive path planning and obstacle avoidance algorithms for dynamic environments.
- To implement a user-friendly application for task assignment and real-time monitoring.
- To evaluate system performance under different indoor scenarios.
- To propose a scalable architecture for future system expansion.

1.6 Scope of the Project

The scope of this project is limited to indoor parcel delivery in structured environments. The system focuses on ground-level navigation in corridors, rooms, and hallways. It includes sensing, navigation, control, and user interface design. Outdoor navigation, stair climbing, elevator integration, and heavy payload transportation are outside the scope of this project. The focus remains on achieving reliable and efficient indoor operation.

1.7 Significance of the Project

This project is significant both academically and practically. From an academic perspective, it contributes to the study of autonomous navigation, robotics, and embedded systems. It provides hands-on experience in integrating hardware and software for real-world applications.

From a practical standpoint, the proposed system offers a cost-effective and scalable solution for indoor parcel delivery. It reduces dependency on manual labor, improves

efficiency, and enhances service reliability. The project also aligns with sustainable development goals by promoting innovation, smart infrastructure, and efficient resource utilization.

1.8 Applications of the Proposed System

1.8.1 Hospitals

The autonomous parcel delivery system can be used to transport medicines, lab samples, medical records, and equipment between departments. This reduces the workload on healthcare staff and minimizes human contact in sensitive areas. Timely delivery improves patient care and operational efficiency. The robot can safely navigate corridors with staff and patients present.

1.8.2 University Campuses

On large campuses, the system can deliver documents, books, and equipment between offices, libraries, and laboratories. It reduces delays caused by long walking distances and limited staff availability. The robot supports efficient inter-departmental operations and helps automate internal logistics. Real-time tracking ensures transparency and accountability.

1.8.3 Office Buildings

Within a corporate setting, the robot is able to take care of the internal mail, files, and small packages. vertical- between departments and floors. This enhances better workflow and minimizes disruption to a minimum. employees. The system provides safe and quality delivery of crucial documents.



Figure 1.1: Example of an autonomous parcel delivery [1]

1.8.4 Industrial Units

The robot is capable of moving tools, components and materials between work stations in factory floors. This minimises the amount of manual operation and accelerates the production. The system promotes the principles of automation and lean manufacturing. It also enhances safety because there is minimization of unnecessary human movement in hazardous areas.

1.8.5 Hotels

In hotels, the system can deliver room service items, laundry, and guest amenities. This enhances guest experience by providing fast and contactless service. The robot operates quietly in hallways and common areas. It also helps reduce staff workload during peak hours.

1.8.6 Emergency and Disaster Zones

In hazardous indoor environments, the robot can deliver essential supplies where human access is risky. It supports rescue operations by carrying tools, food, or medical kits. The system improves safety by reducing human exposure to dangerous conditions. Autonomous navigation ensures reliable operation in unstable environments.

1.9 Limitations of the System

Although the proposed system offers numerous advantages, it has certain limitations. The robot's performance depends on sensor accuracy, battery capacity, and environmental conditions. The system is designed for flat indoor surfaces and may not perform effectively on stairs or uneven terrain. Network connectivity issues may affect real-time monitoring. These limitations provide direction for future improvements.

1.10 Project Plan

The following table shows the proposed project plan and timeline. The entire project, spanning literature review, algorithm development, web interface implementation, and full ROS 2/Gazebo simulation integration, was successfully completed within a total duration of **32 weeks**.

Table 1.2: Proposed Project Plan and Timeline

S#	Tasks	Duration	Resource Person
1	Literature Review	02 Weeks	M.Ali, Aitzaz Arshad, Gul-fam Haider
2	Path Planning (A*/IA*) + Pruning	02 Weeks	M.Ali, Aitzaz Arshad, Gul-fam Haider
3	Path Smoothing (PSO + Cubic Spline)	02 Weeks	M.Ali, Aitzaz Arshad
4	Model Predictive Control (MPC) Design & Implementation	02 Weeks	Aitzaz Arshad, Gulfam Haider
5	Cellular Decomposition & Dynamic Programming Integration	02 Weeks	M.Ali, Aitzaz Arshad, Gul-fam Haider
6	Web Interface – Frontend Development (Flutter)	03 Weeks	M.Ali, Aitzaz Arshad
7	Web Interface – Backend Development (Flask) & API Integration	04 Weeks	M.Ali, Aitzaz Arshad
8	ROS 2 Jazzy Installation & Virtual Environment Configuration	01 Week	M.Ali, Aitzaz Arshad, Gul-fam Haider
9	Gazebo Simulation World Design (Multi-Room Delivery Environment)	02 Weeks	M.Ali, Aitzaz Arshad, Gul-fam Haider
10	SLAM Implementation & Occupancy Grid Mapping	01 Week	M.Ali, Aitzaz Arshad, Gul-fam Haider
11	Porting Path Planning, Smoothing & MPC Algorithms into ROS 2/Gazebo Simulation	03 Weeks	M.Ali, Aitzaz Arshad, Gul-fam Haider
12	Web – Gazebo – ROS 2 Full System Integration	03 Weeks	M.Ali, Aitzaz Arshad
13	System Testing & Performance Evaluation	02 Weeks	Aitzaz Arshad, Gulfam Haider
14	Documentation & Final Report Writing	03 Weeks	M.Ali, Aitzaz Arshad, Gul-fam Haider

1.11 Project Timeline

The following figure shows the Gantt chart for the project schedule, covering the complete 32-week duration of the project.

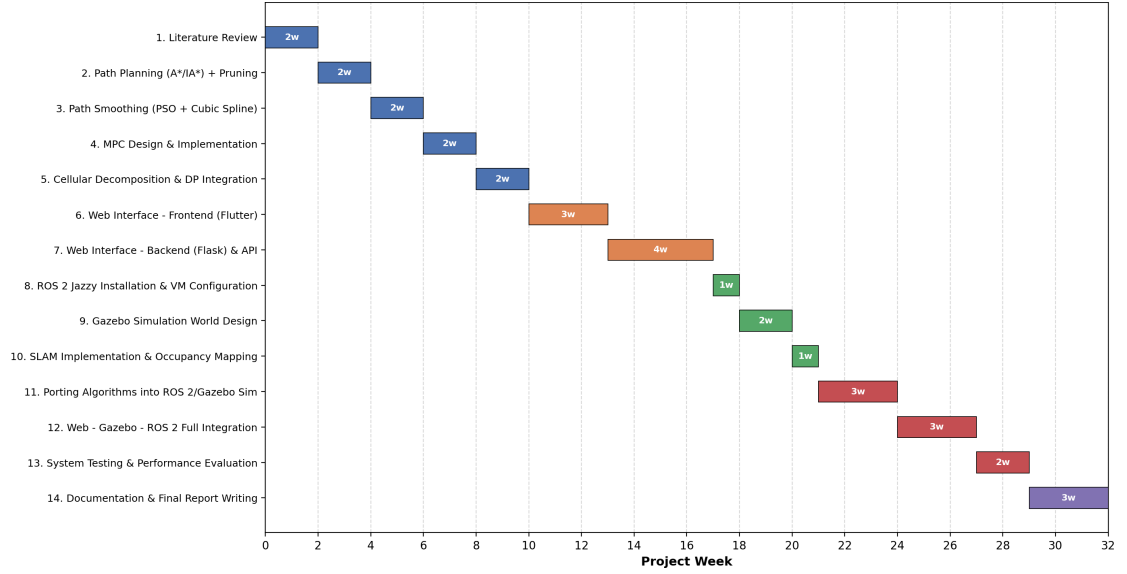


Figure 1.2: Project Timeline (Gantt Chart)

1.11.1 Targeted Sustainable Development Goals

This project aligns with the United Nations Sustainable Development Goals (SDGs) by addressing key challenges in healthcare logistics, education, infrastructure innovation, and sustainable community living. The specific SDGs targeted by the Autonomous Parcel Delivery System are detailed in Table 1.3.

1.12 Report Organization

The report is structured to provide a clear understanding of the project from background to results. Chapter 1 introduces the project, including its motivation, objectives, specifications, and scope. Chapter 2 reviews existing literature on path planning algorithms, mobile robot control strategies, ROS 2 navigation architecture, and previous implementations, highlighting gaps in current approaches. Chapter 3 explains the design and implementation of the custom navigation system, covering the IA* path planning algorithm, path smoothing methods, motion controller design, and ROS 2 integration. Chapter 4 describes the tools and techniques used, including the TurtleBot3 hardware, ROS 2 framework, Gazebo simulation, Python programming, and RViz visualization. Chapter 5 presents project results and evaluation, discussing navigation performance, tracking accuracy, path quality, web interface and Ros. Chapter 6 provides the conclusion and future work, summarizing achievements, lessons learned, limitations, and recommendations for enhancements like dynamic obstacle handling, multi-robot coordination, and real-world deployment. Chapter 7 relates the project to Sustainable Development Goals (SDGs), showing how autonomous navigation contributes to innovation, sustain-

Table 1.3: List of Targeted Sustainable Development Goals

S#	Targeted SDGs	SDG No.	Implementation Detail
1	Good Health and Well-Being	SDG #3	Autonomously transports medical supplies and lab reports in hospitals, reducing staff workload and minimizing human contact to lower cross-contamination risks.
2	Quality Education	SDG #4	Supports academic operations on university campuses by efficiently delivering documents, assignments, and equipment between faculty offices, libraries, and laboratories.
3	Industry, Innovation, and Infrastructure	SDG #9	Develops a scalable autonomous delivery architecture using ROS 2 to modernize internal logistics in warehouses and commercial buildings.
4	Sustainable Cities and Communities	SDG #11	Promotes smart community living in universities and offices by reducing congestion and improving operational efficiency through automated delivery.

able cities, and responsible production. Finally, Chapter 7 provides the conclusion and future work, summarizing achievements, lessons learned, limitations, and recommendations for enhancements like dynamic obstacle handling, multi-robot coordination, and real-world deployment.

1.13 Chapter Summary

This chapter provided a comprehensive introduction to the user-friendly autonomous parcel delivery system. It discussed the background, motivation, problem statement, proposed solution, objectives, scope, significance, applications, limitations, and project milestones. The chapter establishes a strong foundation for the technical and analytical discussions presented in the subsequent chapters.

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